MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **Periodic Test Paper 14 (Cumulative)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

General Instructions:

- This question paper consists of **12 questions** divided into **Five Sections**: A, B, C, D, and E.
- **Section A** comprises **5 questions** of **1 mark** each (Q1–Q4 MCQs, Q5 Assertion-Reason).
- **Section B** comprises **2 questions** (Q6–Q7) of **2 marks** each.
- **Section C** comprises **2 questions** (Q8–Q9) of **3 marks** each.
- **Section D** comprises **2 questions** (Q10–Q11) of **5 marks** each.
- **Section E** comprises **1 Applied Case Study** (Q12) of **5 marks** with internal choice in sub-part (iii).
- All questions are compulsory. Use of calculators is strictly prohibited.

SECTION A: Multiple Choice & Assertion-Reason Questions (5 × 1 = 5 Marks)

Q1. The nth term of an AP is $3n - 1$. Its 12th term is: [1]

(A) 34 (B) 35 (C) 36 (D) 37

Q2. Which term of the AP $3, 15, 27, 39, \ldots$ will be 132 more than its 54th term? [1]

(A) 64th (B) 65th (C) 66th (D) 67th

Q3. The sum of the first 12 terms of the AP $8, 11, 14, 17, \ldots$ is: [1]

(A) 294 (B) 300 (C) 312 (D) 324

Q4. Which term of the AP $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \ldots$ is its first negative term? [1]

(A) 27th (B) 28th (C) 29th (D) 30th

Q5. **Assertion (A):** If the sum of first n terms of an AP is $S_n = 5n - n^2$, then its common difference is -2.
Reason (R): For $S_n = An^2 + Bn$, the common difference is always given by $d = 2A$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.
(B) Both (A) and (R) are true, but (R) is not correct explanation.
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. Find the number of natural numbers between 101 and 999 which are divisible by both 2 and 5. [2]

Q7. If the 8th term of an AP is 31 and the 15th term is 16 more than the 11th term, find the AP. [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. The sum of the 5th and 9th terms of an AP is 72 and the sum of the 7th and 12th terms is 97. Find the AP. [3]

Q9. Find the sum of all three-digit natural numbers which leave remainder 2 when divided by 5. [3]

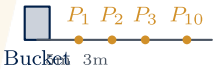
SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q10. If the sum of first m terms of an AP is n and the sum of first n terms is m, show that the sum of its first $(m+n)$ terms is $-(m+n)$. [5]

Q11. A ladder has rungs 25 cm apart. The rungs decrease uniformly in length from 45 cm at the bottom to 25 cm at the top. If the top and bottom rungs are 2.5 m apart, what is the total length of the wood required for the rungs? [5]

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: Sports Day Potato-and-Bucket Relay Race



During an inter-house school sports championship, a potato collection relay race is conducted. A bucket is fixed at the start point.
A straight line of 10 potatoes is laid out such that the first potato is 5 metres from the bucket, and every subsequent potato is placed 3 metres apart. A competitor starts from the bucket, collects the nearest potato, returns to deposit it, and repeats until all 10 potatoes are placed in the bucket.
Based on the above sporting event, answer the following questions:

(i) Find the total distance run by the competitor to collect the first potato and return it to the bucket. [1]

(ii) Find the total round-trip distance run to collect the second potato. [1]

(iii) Calculate the total cumulative distance run by the competitor to collect all 10 potatoes. [3]

OR

(iv) Find the distance covered by the competitor solely during the round trip of collecting the 10th (final) potato. [3]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★